

Next research steps for exact Schrödinger attention

Decision context

The completed evidence does not support scaling the mechanism. Exact amplitude evolution is stable and behaviorally active, but the seed-1701 discovery result did not replicate on the frozen four-seed primary endpoint. D1's restricted temperature-controlled challenge comparison is +0.781 pp pass@32 with a four-seed interval crossing zero, not a reliable difficult-solving benefit. Exact SA also cost about 3.22x softmax in measured training-core time on this reference. No model has been evaluated on the reserved route-policy final-test split.

This document recommends decision gates, not new authorization. Each item says what it can establish and what it cannot. The default scientifically sound option is to stop; every other option requires a new written plan, frozen settings, explicit budget authority, and independent review before execution.

Option 0 — stop and archive (recommended default)

Question. Is current evidence sufficient to justify more experimental expense or claims of an advantage?

Action and endpoint. No additional model execution. Archive the evidence map in the research paper [repo: schrodinger_attention_research_paper.md], the fresh replication report [repo: ../execution/seed_replication/final_report.md], and D1 audit [repo: ../execution/difficult_problem_solving/d1-result-audit.md]. The endpoint is a complete, immutable negative/inconclusive record rather than a new model metric. The unit is the research record.

Cost authority and go/no-go. Zero additional model budget. This is the appropriate no-go if no sponsor is willing to fund a confirmation that is powered for a practically meaningful effect *and* can tolerate the exact method's substantial compute overhead.

Cannot prove. Stopping does not prove the mechanism cannot work elsewhere, with another implementation, at another scale, or on another task. It simply declines to infer a benefit from uncertain, small-scale evidence.

1. D2 runtime forecast before any new model run

Question. Can a fully specified confirmation be executed within a credible, bounded envelope, and what is its smallest useful design?

Minimal test. Perform a read-only engineering forecast using the existing D1 owners, measured core timings, artifact-write timing, memory/profile data, and every required category for a non-overlapping 2,048-problem evaluation. Do not extrapolate a single aggregate D1 duration into a forecast: enumerate model forward/sampling, route verification, cache behavior, output persistence, startup/finalization, audit, and contingency separately. Freeze the exact software revision, CPU/GPU target, model checkpoint set, K, temperatures, panel size, model order, and wall/core-time accounting definitions. Report a range and assumptions rather than a point estimate.

Primary endpoint and unit. The endpoint is a pre-authorized runtime and resource envelope with all cost categories covered; the unit is one complete evaluation owner. A forecast is acceptable only if it identifies what is measured versus assumed and preserves enough budget for independent audit.

Cost-authority gate. D1 explicitly left D2 forecast deferred/no-go because essential scaling categories were missing. This forecast itself should be read-only and must receive a small, separate accounting authorization if it requires profiling. It must not silently run new inference, use final-test models, or change the 7,200-second accounting interpretation. The current qualified operational debit (4764.416447001050/7200 seconds) includes a

300-second administrative uncertainty allowance; it is not a physical-runtime total and should not be combined with historical cumulative totals.

Go/no-go. Go only if a reviewed forecast demonstrates a bounded execution with an explicit scientific effect threshold worth the expected cost. No-go if the forecast remains materially incomplete, exceeds authorized resources, or shows that the design cannot distinguish a benefit commensurate with roughly 3.22x core training cost.

Cannot prove. A forecast cannot show modelling benefit, efficiency on a different accelerator, or a scalable approximation. It only makes a proposed test auditable before resources are spent.

2. Untouched-map confirmation, conditional on a forecast and authority

Question. With all choices frozen before inspection, does SA improve a preselected practical endpoint on genuinely untouched maps?

Minimal test. Reuse the existing four trained seed pairs and their fixed 8,000-update checkpoints; do not train new models in this smallest confirmation. Evaluate a confirmation panel from maps never used for temperature selection, D0/D1 analysis, or checkpoint/model decisions. Preserve the 12x12 route lengths, exact oracle, legal-mask policy, common-uniform rollouts, K=32, map balancing, and paired evaluation protocol unless a separately reviewed change is necessary. Before opening the panel, freeze:

- Primary endpoint: challenge pass@32 at the D1-selected/frozen temperature policy, with a paired-seed estimator.
- Secondary endpoints: Q, U_valid/K, valid_headroom, duplicate and invalid rates, routine Q, and proper scores; label these secondary.
- A minimum practically meaningful effect, decision interval rule, exact seed count, map count, stopping rule, and a single multiplicity policy.
- Temperatures and selection rule. Prefer carrying the D1 rule forward without retuning; if any calibration is necessary, do it only on a distinct tuning panel and never reuse confirmation maps.

The primary unit remains an independent paired seed. Maps support the endpoint but do not multiply the number of training replications. Use a predeclared hierarchical/map-aware uncertainty analysis as a secondary sensitivity check, not pseudoreplication.

Cost-authority gate. First complete Step 1. Then obtain explicit authority for evaluation, verification, and independent audit of these existing checkpoints. The panel must be technically inaccessible to the analysis workflow until the freeze record is accepted. This is a conditional recommendation, not permission to release the existing reserved route-policy final test or generate a substitute favorable panel; that release requires separate explicit authority.

Go/no-go. A positive result requires the primary paired estimate and its predeclared uncertainty/decision rule to clear the practical threshold, while routine quality and cost criteria are not materially worse. If it misses, stop the advantage claim and publish the negative confirmation. If it is ambiguous, do not recursively retune or enlarge the panel without a new plan.

Cannot prove. One confirmation cannot establish language-model scaling, general creativity, mechanism causality, or hardware efficiency. It tests one frozen route-policy claim.

3. Fresh-seed, equal-compute comparison only if powered

Question. Is any quality or challenge-coverage difference worth the compute premium when both architectures receive comparable measured compute rather than equal updates?

Minimal test. Do not run another four-seed equal-update study. First design for the smallest effect that would justify the exact method's overhead. If the confirmation endpoint warrants it, train fresh independent pairs with both an equal-update and an equal-measured-core-compute arm, randomizing or block-balancing model order. Predeclare whether the primary endpoint is compute-to-quality, quality at equal core time, or quality at a fixed update count; it cannot be all three without multiplicity control. Include the full wall/core timing protocol and preserve non-overlapping attempt ownership.

Primary endpoint and unit. The primary unit is a fresh paired seed, with core training seconds (not aggregate ledger debit) as the compute denominator. If route quality at equal compute is primary, report Q and pass@32 at frozen temperature along with Brier/KL and error bars. The observed ratio to beat is about 3.22x core time, not an immutable law of all implementations.

Cost-authority gate. Require a prospective power calculation tied to the minimum practical effect, anticipated paired variance, attrition plan, and budget. Four fresh pairs produced a primary SD of 1.837 pp and a wide interval; no credible result follows from treating 512 route problems as replacement seeds. If the forecasted seed count or budget cannot be funded, choose Option 0.

Go/no-go. Go only when the study is powered for a decision-relevant effect and the exact compute envelope is approved. Claim a practical advantage only if the frozen primary test clears its effect/cost threshold. Otherwise stop; do not select early checkpoints, discovery seed 1701, temperature cells, or challenge subgroups after results.

Cannot prove. Even a powered route-policy result does not establish that phase/evolution is the causal reason or that performance transfers to causal language modeling.

4. Mechanism causal ablations — only after a benefit survives

Question. If a reproducible benefit exists, which element causes it: complex phase, unitary evolution, Hamiltonian symmetrization, optimization path, or extra learned controls?

Minimal test. Do not perform broad ablation fishing now. The existing same-weights dt0 diagnostic already proves that evolution can be active and heterogeneous, while showing no consistent benefit. After a benefit passes an untouched confirmation, compare independently trained, compute-accounted variants with matched capacity and initialization: (a) parameter-matched softmax, (b) nonzero evolution with learned versus zero/fixed/random phase, (c) a separately trained dt=0 control, and, if warranted, a real norm-preserving control. A “phase-only, no-evolution” variant is **not** distinct here: at dt=0, $|\text{sqrt}(\text{softmax}(S)) \cdot \exp(i \text{ phase})|^2$ is exactly softmax(S), regardless of phase. Predefine a small set of contrasts and use the same primary task endpoint as the successful confirmation.

Primary endpoint and unit. A causal contrast is the difference between independently trained variants across fresh paired seeds. Attention TV, eigenphase, and dt magnitude are mechanism measurements, not efficacy endpoints. The primary endpoint is the previously validated benefit metric; secondary mediation-style analyses must be described as associative.

Cost-authority gate. No ablations unless the benefit is both replicated and worth explaining. Require a separate budget because matrix-exponential variants and controls can have different speed profiles.

Go/no-go. Continue only if a predeclared ablation changes the replicated benefit in the predicted direction without an explanatory compute/capacity confound. Otherwise report that behavior differs but mechanism attribution is unresolved.

Cannot prove. Ablations cannot make an internal amplitude analogy into a physical quantum claim, and no dt0 counterfactual alone proves training causality.

5. Separate baseline-representation branch

Question. Are the route benchmark's baseline limits dominated by spatial representation/generalization rather than attention type?

Minimal test. Treat this as a separate baseline research question, not an SA rescue. Use softmax-only models with one prespecified spatial representation alternative (for example, coordinate/graph-aware features) against the current row-token encoder, keeping the oracle, maps, data exposure, optimizer family, and held-out cohorts fixed. Analyze seen pairs, new goals on seen maps, and unseen routine maps without treating their descriptive difference as a causal decomposition. Freeze whether the endpoint is held-out Q, proper score, or a generalization gap before running.

Primary endpoint and unit. Independent softmax seed is the unit. Report training-DAG and aligned held-out DAG proper scores alongside route Q; do not use training loss alone as evidence that the representation solved generalization. The known 8k baseline pattern—training KL improves while mixed held-out KL worsens—motivates this branch but does not identify a cause.

Cost-authority gate. This branch requires its own plan and should not open the final test. It does not authorize new SA training or a post-hoc encoder change within an SA comparison.

Go/no-go. Advance only if the new baseline reliably improves held-out route quality/proper scores at an acceptable cost. If it does, restart any SA comparison from an explicitly new, matched protocol rather than mixing old and new baselines.

Cannot prove. A better representation would not prove softmax is globally superior, nor would it erase the current exact-SA result. It improves the benchmark's diagnostic value.

6. Language-model scaling is last

Question. Does an advantage that survives the preceding gates extend to causal language modeling and longer contexts?

Minimal test. Only after a route-policy confirmation and causal-ablation case, design a new causal formulation rather than reuse the bidirectional symmetric Hamiltonian unchanged. Begin with a numerically verified short-context reference, then compare parameter-, token-, and measured-compute-matched small decoder models across multiple seeds. Predeclare loss/perplexity as primary; include throughput, memory, latency, and causal-mask validation. Exact matrix exponentiation may be a reference only; any approximation requires a separate numerical and fairness validation.

Primary endpoint and unit. Seed-level validation loss/compute-to-target is primary, with task-specific behavior secondary. Use held-out text and avoid interpreting a synthetic route effect as a scaling law.

Cost-authority gate. This is a new project with a new budget, data/license review, hardware plan, and causal-attention design review. It is not covered by the remaining route-study operational envelope.

Go/no-go. Go only after the low-cost evidence shows a practical, replicated, compute-aware benefit. Stop if causal correctness, numerical stability, or compute economics fail before a fair model comparison.

Cannot prove. A small language-model test would still not prove broad scaling behavior; scaling laws require multiple sizes and materially larger budgets.

Recommended sequence

1. Choose Option 0 unless a decision-relevant benefit is worth funding.
2. If funded, perform the read-only D2 forecast and freeze a complete protocol.
3. Run one untouched-map confirmation with settings fixed before access.
4. Only on a positive, practical confirmation, run a powered fresh-seed, equal-compute comparison.
5. Only after benefit survives, investigate causal mechanism; independently improve the baseline representation if needed.
6. Treat causal language scaling as a last, separate programme.

This sequence prevents the common failure mode in exploratory architecture work: turning an interesting seed, subgroup, or internal diagnostic into a reason to retune indefinitely. It protects both possible outcomes—an eventual real effect and a clean, useful negative result.